certAIn

PROJECT INTELLIGENCE

certAIn: AI-Powered Project Planning & Risk Forecasting

Midterm Progress Update — AI PM Capstone Project | Meysameh & Santiago |

AI Planning

From plain-language brief to structured task generation

DAG Workflow Modeling

Dependency graphs with critical-path analysis

Probabilistic Forecasting

Monte Carlo simulation for schedule uncertainty

Risk Monitoring

ML-based risk classification & real-time alerts

certAIn takes you from a plain-language project brief or project file to a — all in one platform.

Problem & Motivation

Why This Project Matters

Spreadsheet-Driven Planning

Many project teams still plan with **spreadsheets, static timelines, and manual assumptions**

No Uncertainty Modeling

Those methods do not model uncertainty, dependency effects, or evolving risk very well

Unrealistic Deadlines

Teams often commit to deadlines without a realistic view of schedule risk

Stakeholders Underserved

Stakeholders need clearer, more data-driven planning support

certAIn turns static planning into probabilistic, AI-assisted decision support.

What We've Built So Far

Streamlit App

A working Streamlit application using app1.py

SaaS Interface

SaaS-style interface for presentation plus dashboard workflow for analysis

AI Task Generation

AI-assisted task generation from a project description

DAG & Critical Path

DAG-based dependency modeling & critical-path analysis

Monte Carlo Forecasting

Monte Carlo forecasting

ML Risk Classification

Advisory ML-based risk classification

Scenario & Summary

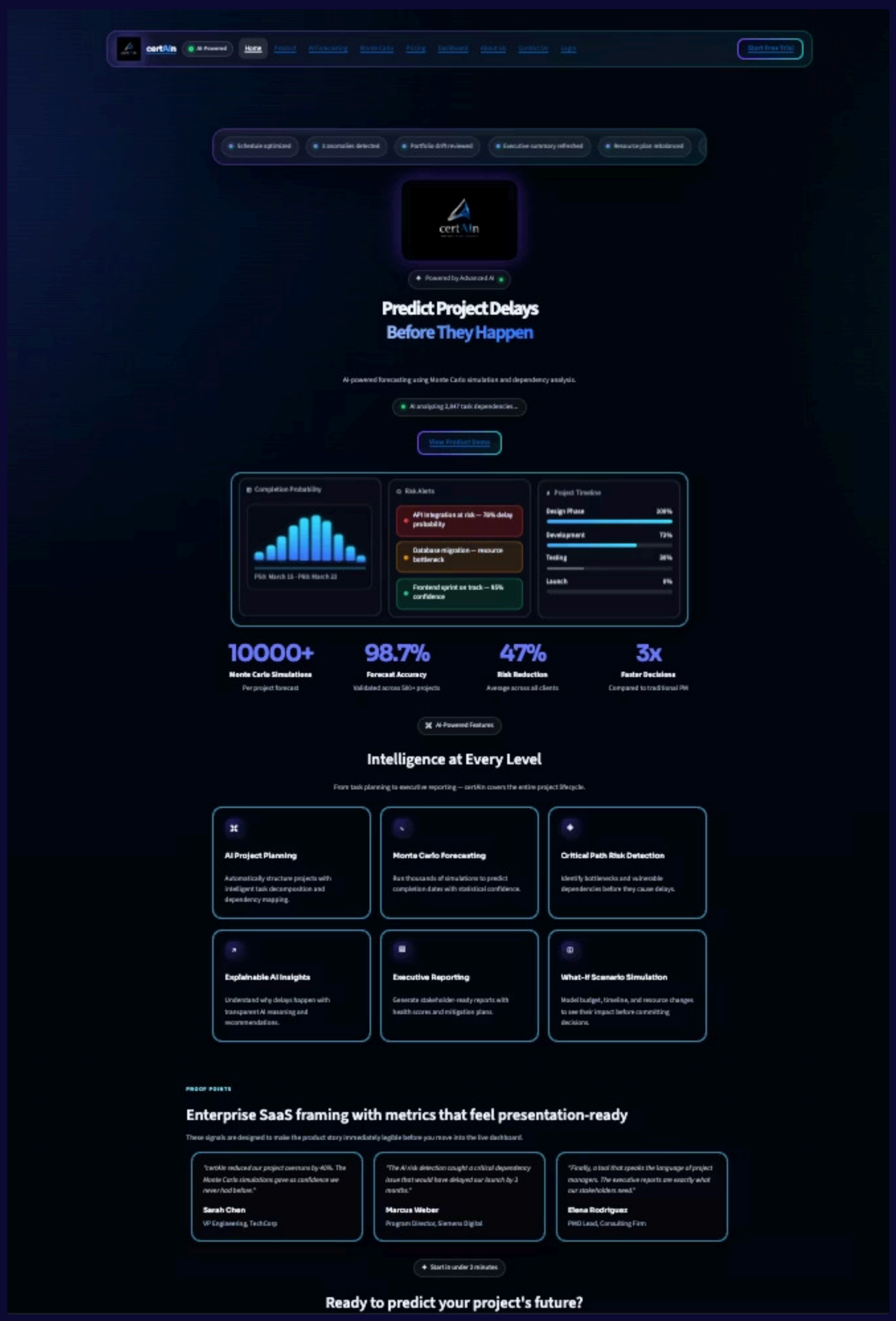
Scenario analysis, monitoring view, & executive summary outputs

Session Persistence

SQLite-backed session persistence

User-Facing Pages

- marketing/product pages
- dashboard
- user guide
- decision hub
- AI task architect
- risk intelligence
- Monte Carlo simulator
- what-if scenarios
- executive summary



End-to-End Workflow

User enters a plain-language project description / uploads a project file

LLM generates structured tasks

Tasks are converted into a dependency graph

Critical path is computed

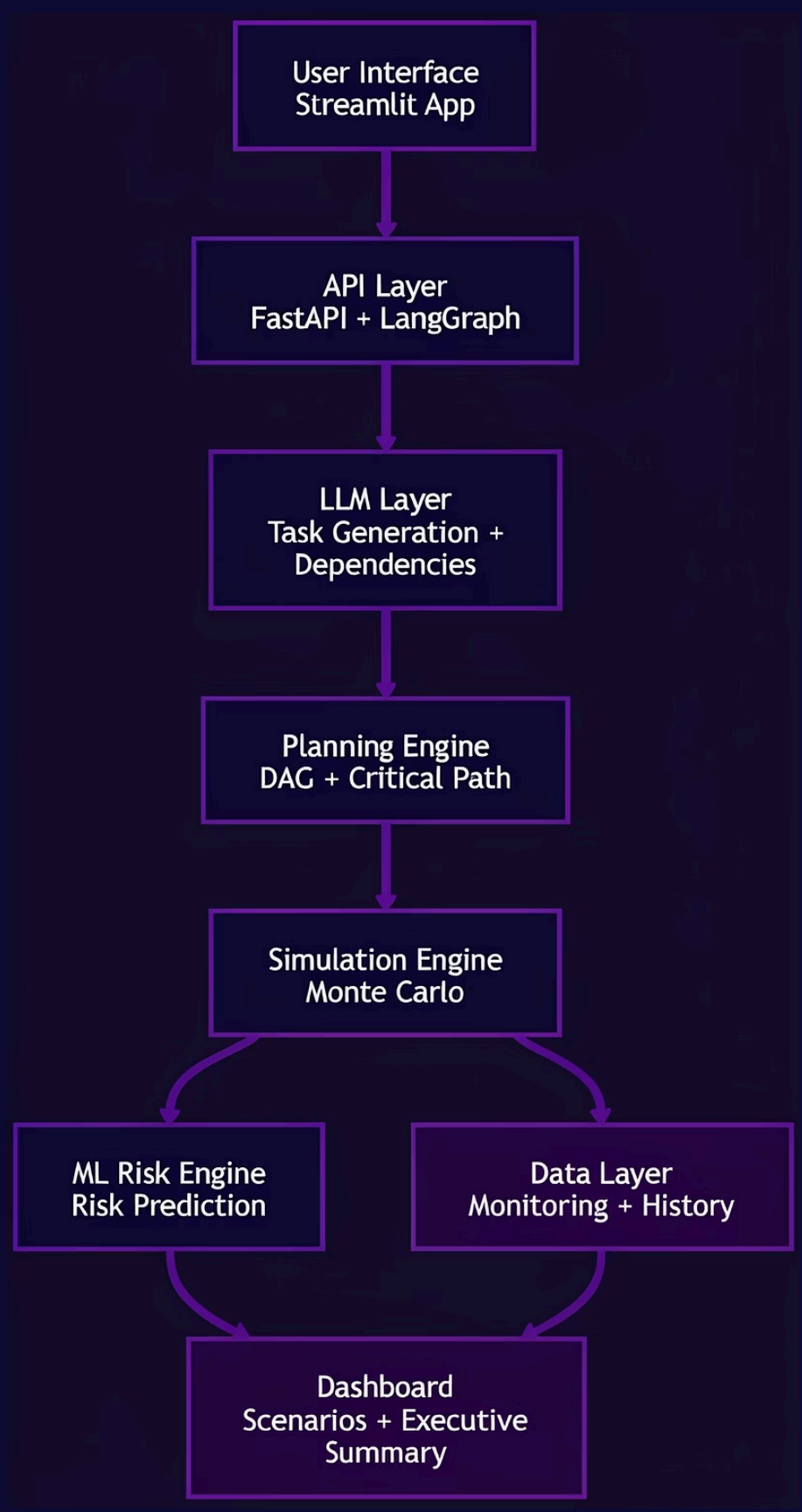
Monte Carlo simulation estimates completion uncertainty

ML classifier adds task-level risk labels

Dashboard presents risk, scenarios, & executive summary

The Pipeline

The platform is not just a dashboard. It is a pipeline: text input → structured planning → graph analysis → simulation → ML risk support.



Data Assets Used

Runtime Data Files (used by the app)

project_portfolio_history.csv

360 rows, 24 columns

risk_monitoring_snapshot.csv

270 rows, 12 columns

Saved model artifact

risk_classifier.joblib

Saved metrics file

risk_model_metrics.json

What These Support

- portfolio context in the dashboard
- monitoring and post-deployment drift view
- advisory task-level risk scoring

Dataset Training results

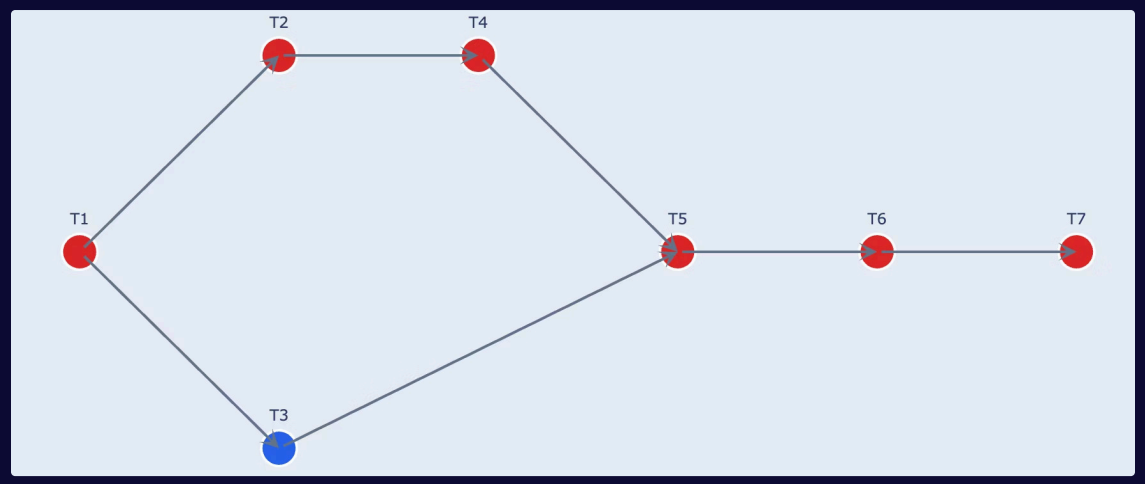
7.86 days	15.2 days
Avg. Historical Delay	Avg. Forecast Error
77.41%	64.31%
Monitoring Match Rate	Avg. Prediction Confidence
0.383	10
Avg. Drift Score	Open Monitoring Alerts

The risk model artifact was trained on a task-level dataset referenced in the metrics file, reflecting real-world project characteristics.

DAG Modeling & Forecasting

Why DAG Modeling Matters

- Tasks are modeled as nodes
- Dependencies are modeled as directed edges
- The graph must be acyclic, so there are no circular dependencies
- This allows us to identify the critical path and simulate completion under uncertainty



DAG modeling is the bridge between generated tasks and realistic schedule forecasting.

Where It's Used in the Project

Graph construction & validation

src/modeling/graph_builder.py

Critical-path calculation

src/modeling/critical_path.py

Simulation on the graph

`src/simulation/monte_carlo.py`

Dependency graph visualization

`src/visualization/charts.py`

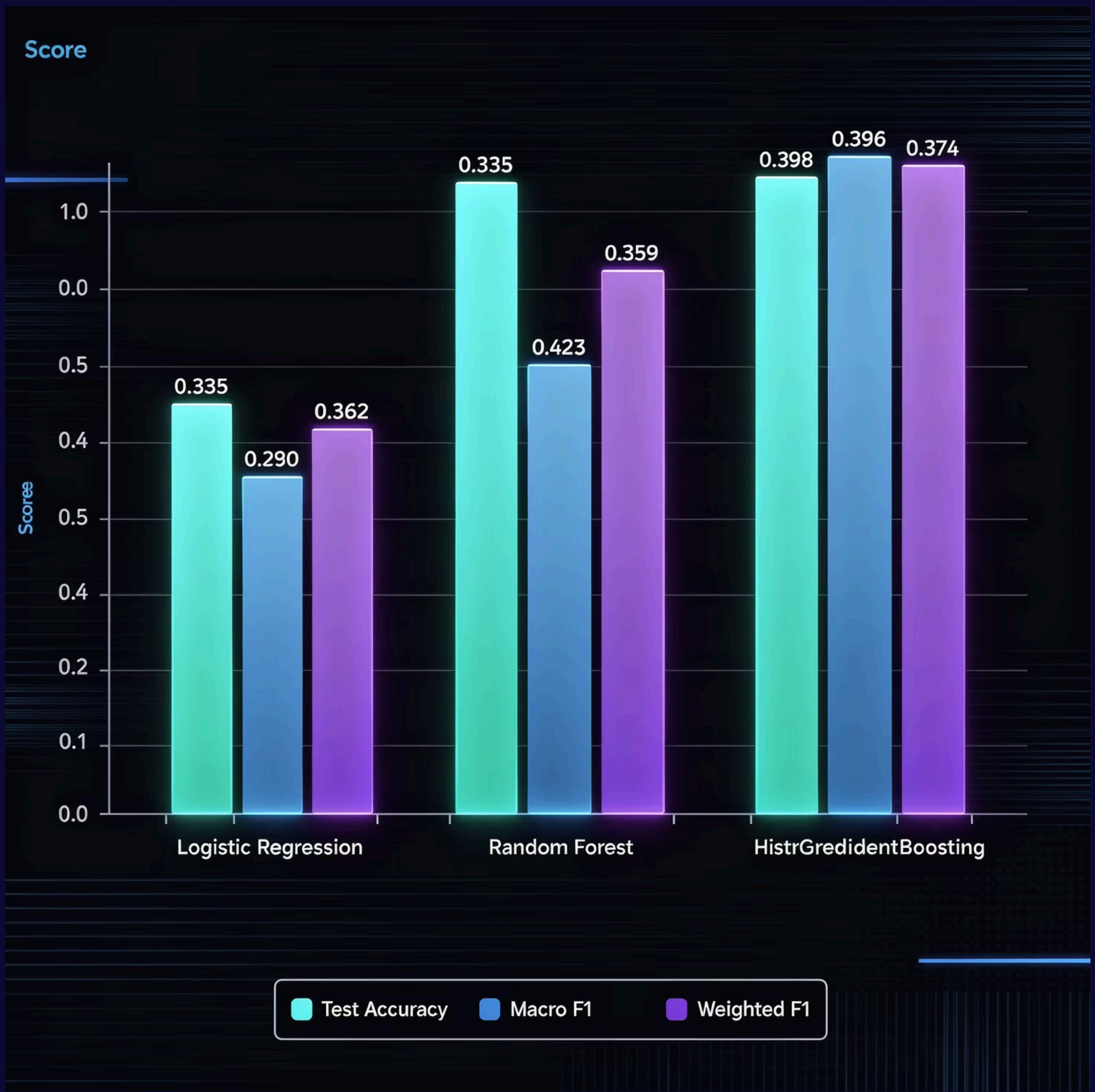
Runtime use

app1.py

Baseline & Model Comparison

We compared multiple models and used Logistic Regression as our baseline.

Model Comparison:



Key Findings:

- **Logistic Regression** gave us a simple and interpretable baseline.
- Tree-based ensemble models (**Random Forest** and **Extra Trees**) capture nonlinear relationships and improve performance.
- **HistGradientBoosting** achieved the strongest overall performance based on cross-validated Macro F1 and was selected as the final model.

Ensemble methods clearly outperformed the baseline model.

Current Model Status

Primary Model: HistGradientBoosting

Why: strong accuracy, strong class performance, and robust behavior on structured tabular data.

Current App Artifact at Runtime

- Saved advisory model artifact: `models/risk_classifier.joblib`
- Saved metrics file currently named: `HistGradientBoosting` (app1.py)
- Model version: `v1-advisory-multimodel` (config.py)

Top Features (schema.py)

1. `Labor_Required`
2. `Equipment_Units`
3. `Resource_Constraint_Score`

Interpretation

`Logistic Regression` baseline, compared against `Random Forest`, Extra Trees, and HistGradientBoosting.

The current Streamlit app still loads this saved artifact & surfaces its model name, version, & top features in the dashboard.

HistGradientBoosting — Performance Metrics

Current Progress Evidence

The following deliverables are already complete and working:

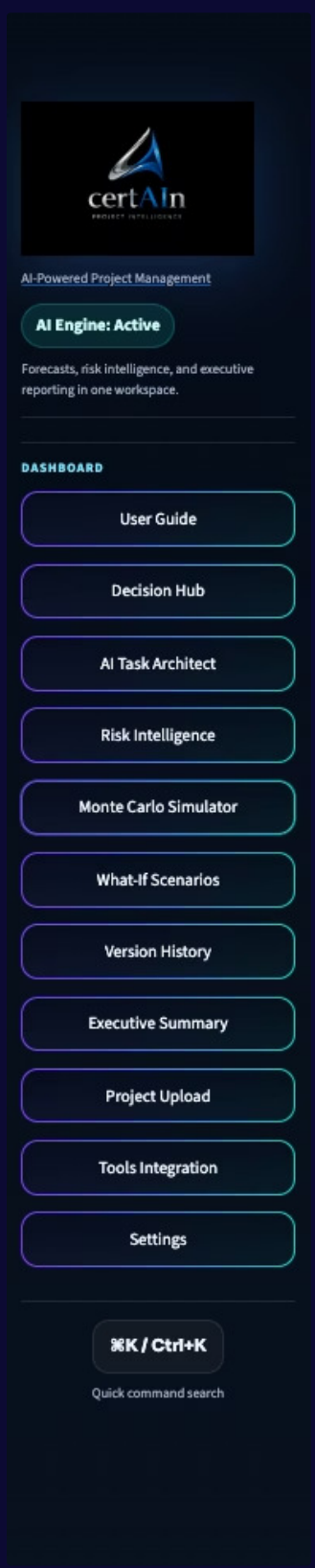
5 Notebooks

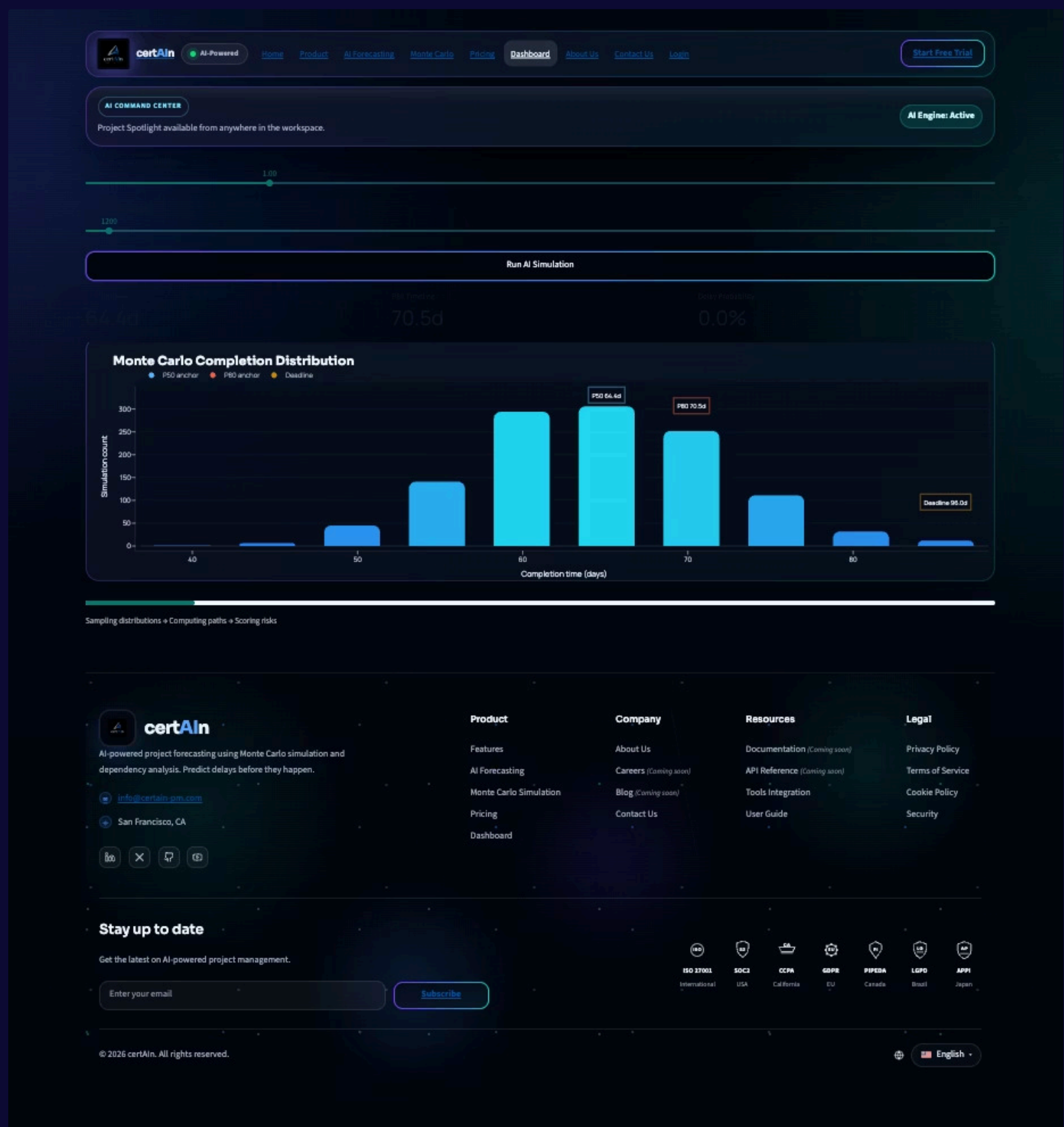
- Business understanding
- EDA
- Baseline & model comparison
- Risk forecasting DAG & Monte Carlo
- Model monitoring

Streamlit MVP

- Current Streamlit MVP in `app1.py`
- Saved model artifact & metrics file
- Portfolio & monitoring data integrated into the app
- Dependency graph visualization
- Scenario comparison
- Session history in SQLite
- 9 automated test files across core modules (tests)

> Evidence covers: technical EDA, multiple model comparison, dependency-aware simulation, model monitoring, model deployment, and live dashboard progress.





Limitations & Next Steps

What still needs improvement

Current Limitations

- ML classifier is task-level; dashboard is project-level — alignment needed
- App runs locally; final deployment/hosting — to be finalized
- LLM task generation still in mock mode — to be activated in deployment

Next Steps before final presentation

- Improve model quality & calibration
- Align task-level & project-level analytics
- Finalize deployment setup
- Package final screenshots, slides, & demo flow in the repo
- Strengthen LLM → DAG → simulation → monitoring integration

The core system works — the final stage is about tightening integration and polishing delivery.

Thank You

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